KAIZEN MATHS

Lesson starter

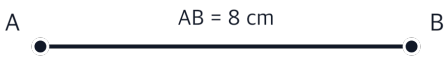
Loci and Constructions

Constructions

Name _____

SA1

Construct each perpendicular bisector using compass arcs from both endpoints.



Construct the perpendicular bisector of AB, where AB = 8 cm.

Show your working

KAIZEN MATHS

Lesson starter

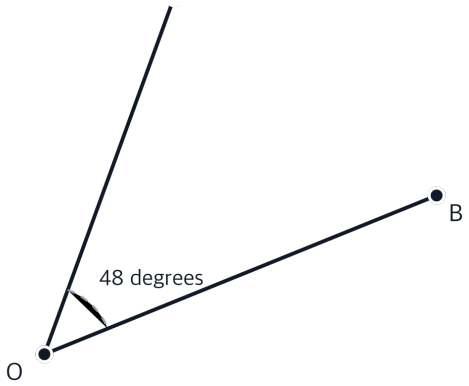
Loci and Constructions

Constructions

Name _____

SB1

Construct each angle bisector using equal compass arcs from the two arms of the angle.



Construct the angle bisector of angle AOB.

Show your working

KAIZEN MATHS

Lesson starter

Loci and Constructions

Constructions

Name _____

SC1

Construct a perpendicular line through the given point on the line.



Construct a perpendicular to line l through point P.

GEO-F01A-L1 · SC

Show your working

KAIZEN MATHS

Lesson starter

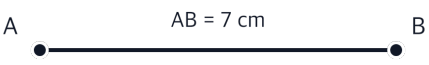
Loci and Constructions

Constructions

Name _____

SD1

Construct each triangle from the three side lengths given.



Construct triangle ABC with AB = 7 cm, AC = 5 cm, and BC = 7 cm.

GEO-F01A-L1 · SD

Show your working

KAIZEN MATHS

Exit ticket

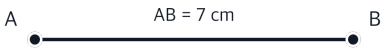
Loci and Constructions

Constructions

Name _____

EA1

Construct each perpendicular bisector using compass arcs from both endpoints.



Construct the perpendicular bisector of AB, where AB = 7 cm.

Show your working

KAIZEN MATHS

Exit ticket

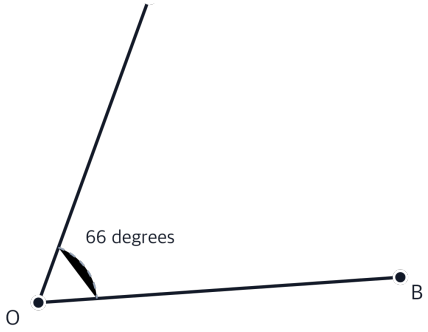
Loci and Constructions

Constructions

Name _____

EB1

Construct each angle bisector using equal compass arcs from the two arms of the angle.



Construct the angle bisector of angle AOB.

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Constructions

Name _____

EA1

Construct each perpendicular bisector using compass arcs from both endpoints.

A
AB = 7 cm
B

Construct the perpendicular bisector of AB, where AB = 7 cm.

GEO-F01A-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Constructions

Name _____

EB1

Construct each angle bisector using equal compass arcs from the two arms of the angle.

66 degrees
O
B

Construct the angle bisector of angle AOB.

GEO-F01A-L1 · EB

Show your working

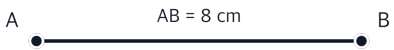
Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Construct each perpendicular bisector using compass arcs from both endpoints.



Construct the perpendicular bisector of AB, where AB = 8 cm.

Answer:

The perpendicular bisector of AB.

Working:

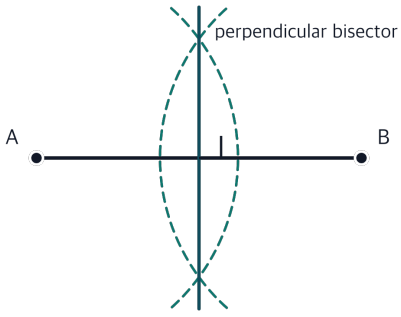
Set the compass radius to more than half the length of AB.

With centre A, draw arcs above and below the line segment.

Keep the same compass radius. With centre B, draw arcs to cross the first pair of arcs.

Use a ruler to join the two arc intersection points.

The line you have drawn meets AB at 90 degrees and cuts AB into two equal parts.



Every point on this perpendicular bisector is equidistant from A and B.

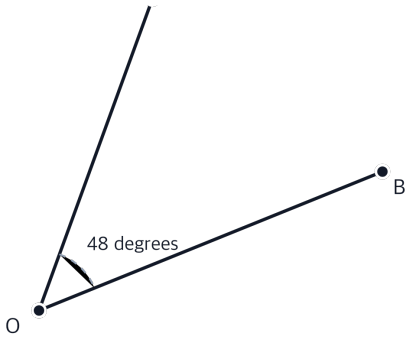
Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Construct each angle bisector using equal compass arcs from the two arms of the angle.



Construct the angle bisector of angle AOB.

Answer:

The ray from O that bisects angle AOB.

Working:

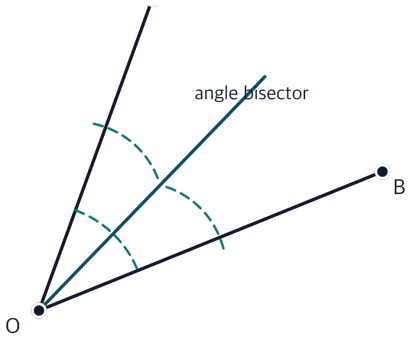
Place the compass point on O and draw an arc that cuts both arms of the angle.

Label the two points where the arc meets the arms.

Keeping the compass width fixed, draw an arc from each of those two points inside the angle.

The two new arcs cross inside the angle.

Use a ruler to join O to that crossing point.



The new ray divides angle AOB into two equal angles.

Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Construct a perpendicular line through the given point on the line.



Construct a perpendicular to line l through point P.

Answer:

A line through P at 90 degrees to line l.

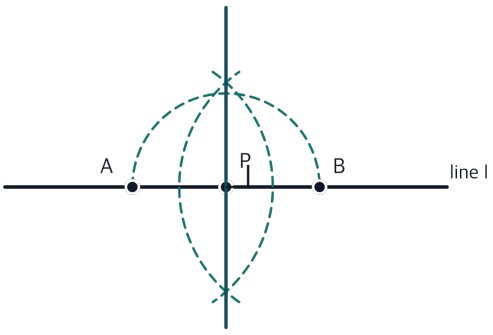
Working:

With centre P, draw an arc that cuts line l at two points.

Call those points A and B.

Using the same compass width from A and B, draw arcs above the line so that they cross.

Use a ruler to join P to the arc crossing point.



The new line is perpendicular to line l because it is the perpendicular bisector of AB.

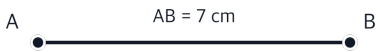
Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Construct each triangle from the three side lengths given.



Construct triangle ABC with AB = 7 cm, AC = 5 cm, and BC = 7 cm.

Answer:

Triangle ABC constructed from the three side lengths.

Working:

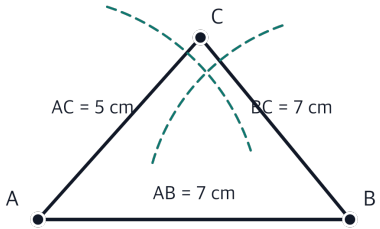
Draw the base AB = 7 cm.

Set the compass width to 5 cm. With centre A, draw an arc above AB.

Set the compass width to 7 cm. With centre B, draw an arc to cross the first arc.

Label the arc intersection C.

Use a ruler to join A to C and B to C.



The completed triangle has the required three side lengths.

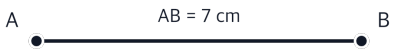
Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Construct each perpendicular bisector using compass arcs from both endpoints.



Construct the perpendicular bisector of AB, where AB = 7 cm.

Answer:

The perpendicular bisector of AB.

Working:

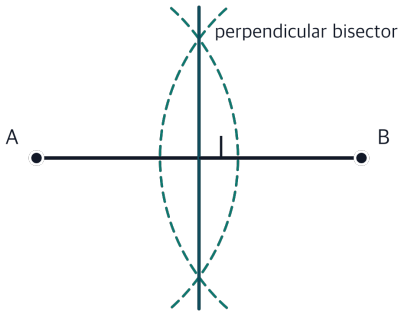
Set the compass radius to more than half the length of AB.

With centre A, draw arcs above and below the line segment.

Keep the same compass radius. With centre B, draw arcs to cross the first pair of arcs.

Use a ruler to join the two arc intersection points.

The line you have drawn meets AB at 90 degrees and cuts AB into two equal parts.



Every point on this perpendicular bisector is equidistant from A and B.

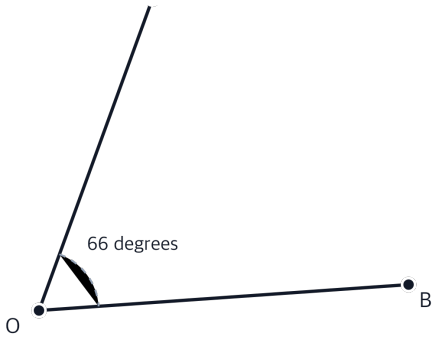
Teacher answers & working

Loci and Constructions · Constructions

GEO-F01A-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Construct each angle bisector using equal compass arcs from the two arms of the angle.



Construct the angle bisector of angle AOB.

Answer:

The ray from O that bisects angle AOB.

Working:

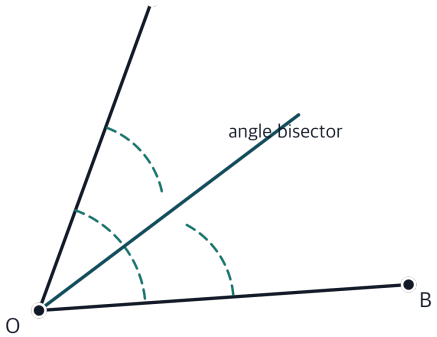
Place the compass point on O and draw an arc that cuts both arms of the angle.

Label the two points where the arc meets the arms.

Keeping the compass width fixed, draw an arc from each of those two points inside the angle.

The two new arcs cross inside the angle.

Use a ruler to join O to that crossing point.



The new ray divides angle AOB into two equal angles.